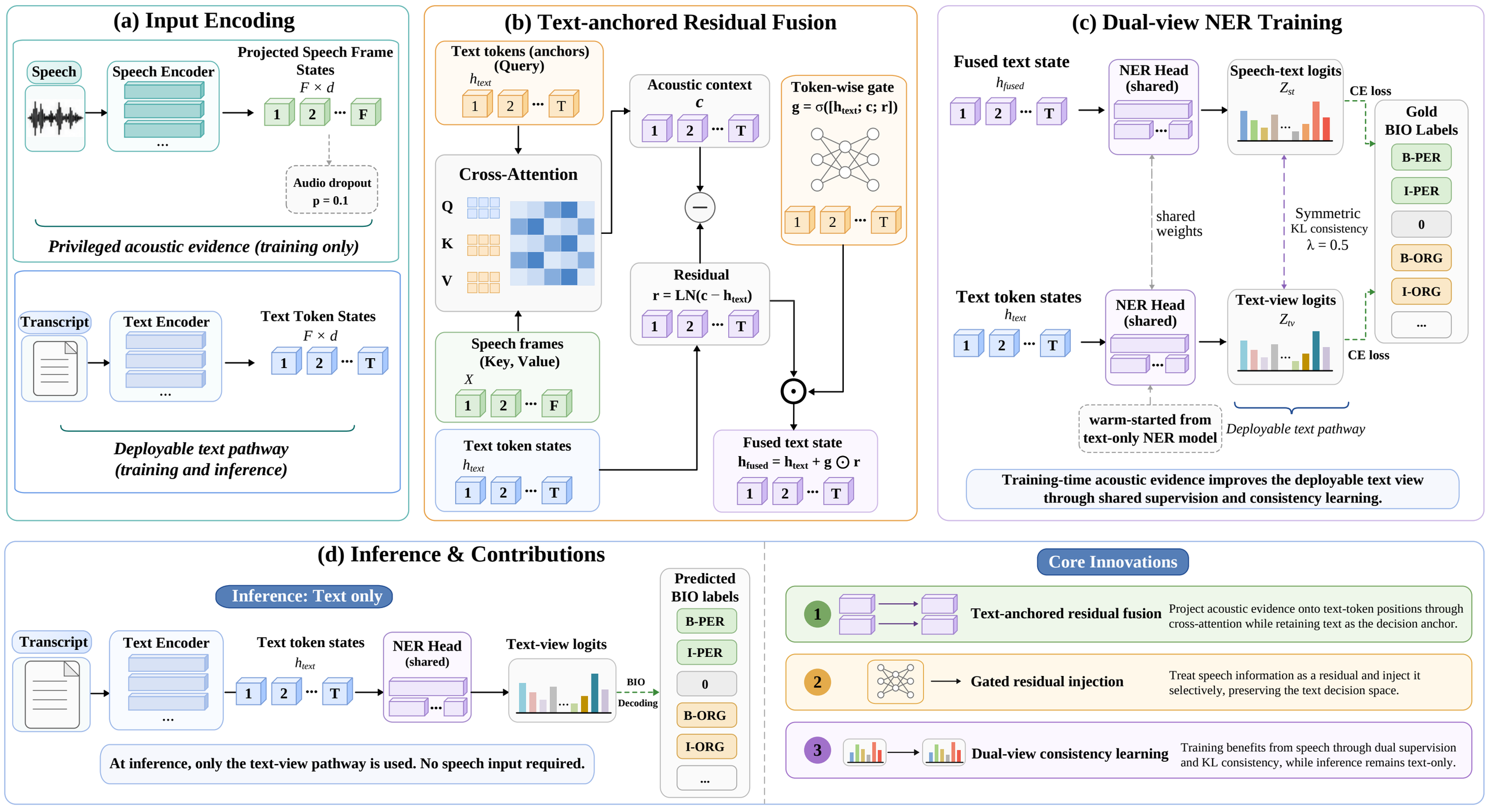

(a) Input Encoding
Speech
Speech Encoder
Projected Speech Frame States
F × d
Audio dropout
p = 0.1
Privileged acoustic evidence (training only)
Transcript
Text Encoder
Text Token States
F × d
Deployable text pathway
(training and inference)
(b) Text-anchored Residual Fusion
Text tokens (anchors)
(Query)
Cross-Attention
Q
K
V
Speech frames
(Key, Value)
X
Text token states
Acoustic context
c
Residual
r = LN(c − h_text)
Token-wise gate
g = σ([h_text; c; r])
Fused text state
h_fused = h_text + g ⊙ r
(c) Dual-view NER Training
Fused text state
h_fused
NER Head
(shared)
Speech-text logits
Z_st
CE loss
Gold
BIO Labels
B-PER
I-PER
0
B-ORG
I-ORG
shared
weights
Symmetric
KL consistency
λ = 0.5
Text token states
h_text
NER Head
(shared)
Text-view logits
Z_tv
CE loss
warm-started from
text-only NER model
Deployable text pathway
Training-time acoustic evidence improves the deployable text view
through shared supervision and consistency learning.
(d) Inference & Contributions
Inference: Text only
Transcript
Text Encoder
Text token states
h_text
NER Head
(shared)
Text-view logits
BIO
Decoding
Predicted
BIO labels
B-PER
I-PER
0
B-ORG
I-ORG
At inference, only the text-view pathway is used. No speech input required.
Core Innovations
1
Text-anchored residual fusion
Project acoustic evidence onto text-token positions through cross-attention while retaining text as the decision anchor.
2
Gated residual injection
Treat speech information as a residual and inject it selectively, preserving the text decision space.
3
Dual-view consistency learning
Training benefits from speech through dual supervision and KL consistency, while inference remains text-only.